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Unintended consequences of water conservation on the use of treated municipal wastewater

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Municipal water managers are intensifying efforts to reduce urban water use and increase the reliability of local water supplies to combat rising water scarcity and drought. Incentivizing increases in water-use efficiency and mandating conservation are two strategies to rein in demand. Concurrently, local water supplies are being augmented through investments in reclaiming and treating municipal wastewater. Although reducing urban water consumption is necessary to deal with population growth and a more variable climate, it does come at a cost. In particular, cutting back on indoor water consumption impacts the generation and quality of wastewater, which can have widespread and underappreciated consequences on human society and the environment. Here, to quantify these impacts, we tracked monthly effluent flow, salinity levels and the properties of 34 wastewater treatment plants throughout Southern California from 2013 to 2017—a period that included extreme drought and abundant precipitation. Our analysis demonstrates that conservation measures significantly reduced effluent flow and increased effluent salinity (P value ≤ 0.05). Our findings further highlight the need for policymakers to recognize the interdependencies and complexities within a water system.

As cities in arid and semi-arid regions worldwide confront a future consisting of growing populations, less reliable water supplies and more intense and frequent droughts, municipal water managers are expanding their portfolio of water management strategies to both reduce demand and augment supplies[1–3]. Efforts to reduce demand, which addresses one-half of the water scarcity puzzle, have given rise to an explosion of price and non-price approaches that are intended to increase water-use efficiency and water conservation, including requiring lower-water-use technologies (for example, low-flow toilets and showerheads) in new constructions, encouraging and incentivizing the adoption of such technologies in older buildings and homes, using non-price approaches such as social comparisons, and developing programs and pricing structures to encourage reductions in outdoor water use[4–12]. Similar to what has been observed in irrigated agriculture, these efforts have led to increases in water-use efficiency and a levelling off of water use in many municipalities worldwide. For example, between 1995 and 2010, daily average water use per capita in California's urban sector decreased by nearly 30%, which was largely due to programs that have encouraged, required or incentivized the adoption of lower-water-use technologies[13].

On the supply side of the ledger, municipalities are increasingly augmenting their water-supply portfolios with treated municipal wastewater (TMW). In the United States, municipal wastewater reuse—which supplies agricultural and urban irrigation, industrial processes and potable water use—increased from 4 million $m^3\,d^{-1}$ in 1995 to nearly 8.5 million $m^3\,d^{-1}$ in 2015 (ref. [14]). In California, reuse of TMW for urban uses increased from 34 million m^3 to more than 313 million m^3 annually from 1970 to 2015, even outpacing agricultural reuse[15]. Owing to the cost and reliability advantages that are associated with TMW, such trends will probably continue[16,17]. The National Research Council estimates that TMW has the potential to provide up to 27% of the overall water supplies of US municipalities, up from its current 5% (ref. [18]). California alone plans to increase recycled water use by nearly 500%—from 521 million m^3 to more than 3,083 million m^3 per year —by 2029 (ref. [15]).

However, a more careful examination of the potential synergistic effects among some types of conservation and reuse exposes a conflict. By invoking basic water-balance accounting[19,20], it is clear that indoor water conservation may reduce the quantity and quality of wastewater that is available for reuse (Fig. 1). The extent to which reductions in water use impact the generation and quality of wastewater has yet to be explored systematically, except for one-off analyses of single plants[14]. The potential impacts of declining flows within an interconnected water system can be broad, including reducing the availability of treated wastewater for irrigated agriculture, golf courses, groundwater recharge or critical flows to local effluent-dominated streams and rivers that provide habitat and recreational services. Owing to the coevolution and interdependencies of factors that comprise our complex human water systems[21–25], such impacts can even result in increased water scarcity and necessitate additional water reduction requests and measures.

To demonstrate the potential impacts of water conservation on the availability and quality of wastewater, we developed a monthly dataset from January 2013 to December 2017, tracking effluent flow and salinity levels in 33 wastewater treatment plants (WWTPs) throughout Southern California as well as the characteristics of the corresponding water agencies (that is, gallons per capita per day, conservation achievements, total potable water production and the percentage of residential water use). This unique dataset enables us to include fixed effects of the WWTPs and address possible bias in our statistical analysis that might arise from omitted time-invariant factors. Our analysis demonstrates that, compared with the baseline levels, conservation measures caused a significant decrease in effluent flow and an increase in effluent salinity. There is also evidence that not all of the conservation policies impact effluent quantity and quality alike, and that changes in source-water salinity may compound the challenges. Although it is clear that municipalities will

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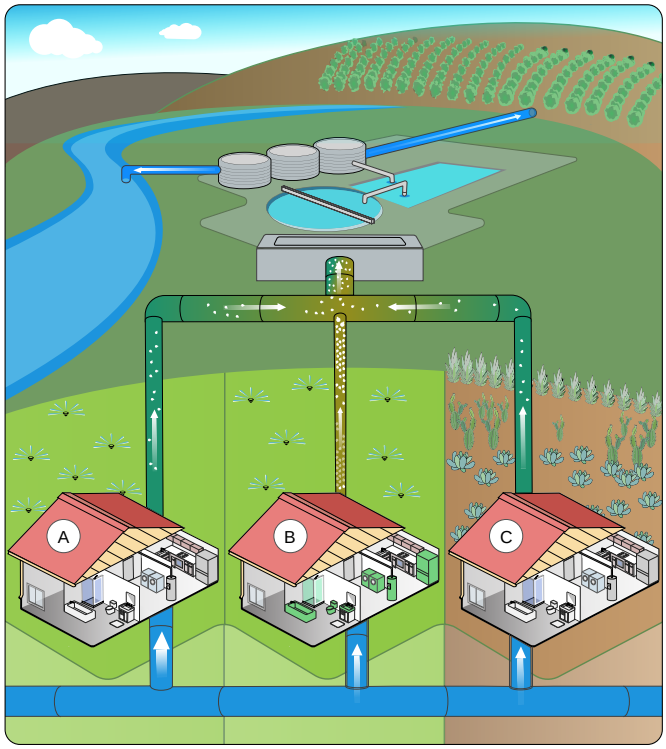


Fig. 1 | Practical examples of water conservation in municipalities. Inefficient indoor and outdoor water use (example A); efficient indoor but inefficient outdoor use (example B); and inefficient indoor but efficient outdoor use (example C). We assume that example A uses the highest amount of water among the three houses. For examples B and C, water usage is equal but lower than the water usage of example A (water use, A > B = C). Examples A and C have the same effluent flow volume but a greater effluent flow volume than example B (effluent flow volume, B < A = C). Examples A and C would have the same effluent flow salinity levels but lower effluent flow salinity levels than example B (effluent flow salinity, B > A = C).

need to continue to invest in improving water-use efficiency and impose restrictions on water use or implement conservation strategies (for example, imposing restrictions on water use) to deal with long-term scarcity and short-term shortages[26], respectively, a better understanding of how such investments and restrictions impede wastewater reuse can inform more targeted and efficient policies that consider the inherent linkages within water systems such that unintended and inefficient conflicts are minimized.

Drought policy, water conservation and wastewater effluent

During the period of 2012–2016, the state of California enacted a series of statewide drought policies in response to one of its hottest and driest droughts on record[27,28]. Whereas agricultural water users—who use approximately 40% of the annual water supply in California, yet account for only approximately 16% of water use in Southern California—had deliveries curtailed between 0% and 100% depending on the seniority of their water rights, municipal water users, who comprise 10% of the water use annually in California but 80% in Southern California, confronted three distinct water conservation requests from the state water board[13,29,30]. In June 2014, the state made a voluntary request for all public water agencies to reduce water use by 20% relative to a 2013 baseline. As the drought lingered into 2015, the state then imposed a conservation mandate in June 2015 that required public water agencies to reduce their water use relative to 2013 by a percentage varying from 4% to 36% depending on each agency's GPCD water use in 2013 to achieve an overall 25% reduction in statewide urban water use, approximately 1.3 million acre-feet (1.6 billion m^3)[31,32]. With pushback from agencies who were following their drought contingency plans and felt their situation did not warrant such draconian reductions, coupled with a slight reprieve from the drought in early 2016, the state rescinded the conservation mandate in June 2016 and replaced it with a self-certification period that lasted until May 2017. This self-certification period required local suppliers to apply a prescribed reliability stress test and set water savings targets on the basis of anticipated water shortfalls over the next three years. Over this same period in which the state was enacting these different statewide conservation policies, voters passed a bond measure that included more than US$725 million for water recycling projects[33].

Figure 2 shows the locations of our sample of 34 WWTPs throughout Southern California, along with the average percentage changes in GPCD relative to a baseline of 2013 for each of the three—voluntary, mandatory and self-certification—conservation policy periods discussed above. The GPCD estimates are for single-family residential accounts from the water districts that deliver potable water that is transformed into wastewater through municipal uses and delivered to WWTPs for treatment. We focus on residential water use because it comprises the bulk of urban water use (67%), significantly more than commercial and institutional use (25%) and industrial water use (6%). For this map (Fig. 2), we used 2013 monthly data as a baseline and, for each WWTP, calculated the weighted average percentage change in monthly GPCD for each conservation policy. Negative values indicate reductions compared with the 2013 levels; positive values, which were very rare, indicate increases relative to the 2013 levels. The brighter the colour (for example, going from green to yellow to red), the greater the percentage reduction in GPCD relative to 2013.

As shown in Fig. 2, single-family residential water use changed during each of the three periods relative to the 2013 levels, but more so under the conservation mandate. During the voluntary reduction period, the WWTPs in our sample were receiving influent from water districts that were experiencing a 6.56% reduction in single-family residential water demands on average. However, under the conservation mandate, that average dropped to slightly more than 20% (consistent with the state achieving its desired goal of a 25% statewide reduction). During the self-certification period, average water use by single-family residential accounts dropped by 17.04%. Although this map does not provide information on wastewater effluent flows during this period, if conservation was primarily achieved from reductions in outdoor water use, there should not be any appreciable change in effluent flow volume or quality. However, if the reduction targets were achieved through indoor conservation, one would expect to see decreased effluent flows along with increased levels of influent salinity (due to the increased concentration of total dissolved solids (TDS) and other constituents in the wastewater from less indoor water use). Unless advanced treatment processes are in place, such as reverse osmosis (RO), the influent salinity concentrations will continue through to the effluent. Although advanced treatment costs, such as RO, for treating wastewater are generally lower than importing water and use approximately one-third to one-half of the energy compared with ocean-water desalination, the capital costs are substantial, as are the challenges that are associated with the disposal of the brine that is generated[18,34].

Water conservation and WWTP effluent flow and quality

To understand how water conservation impacted WWTP effluent flow and quality during the drought, we plotted the monthly percentage change in single-family residential GPCD, effluent flow and

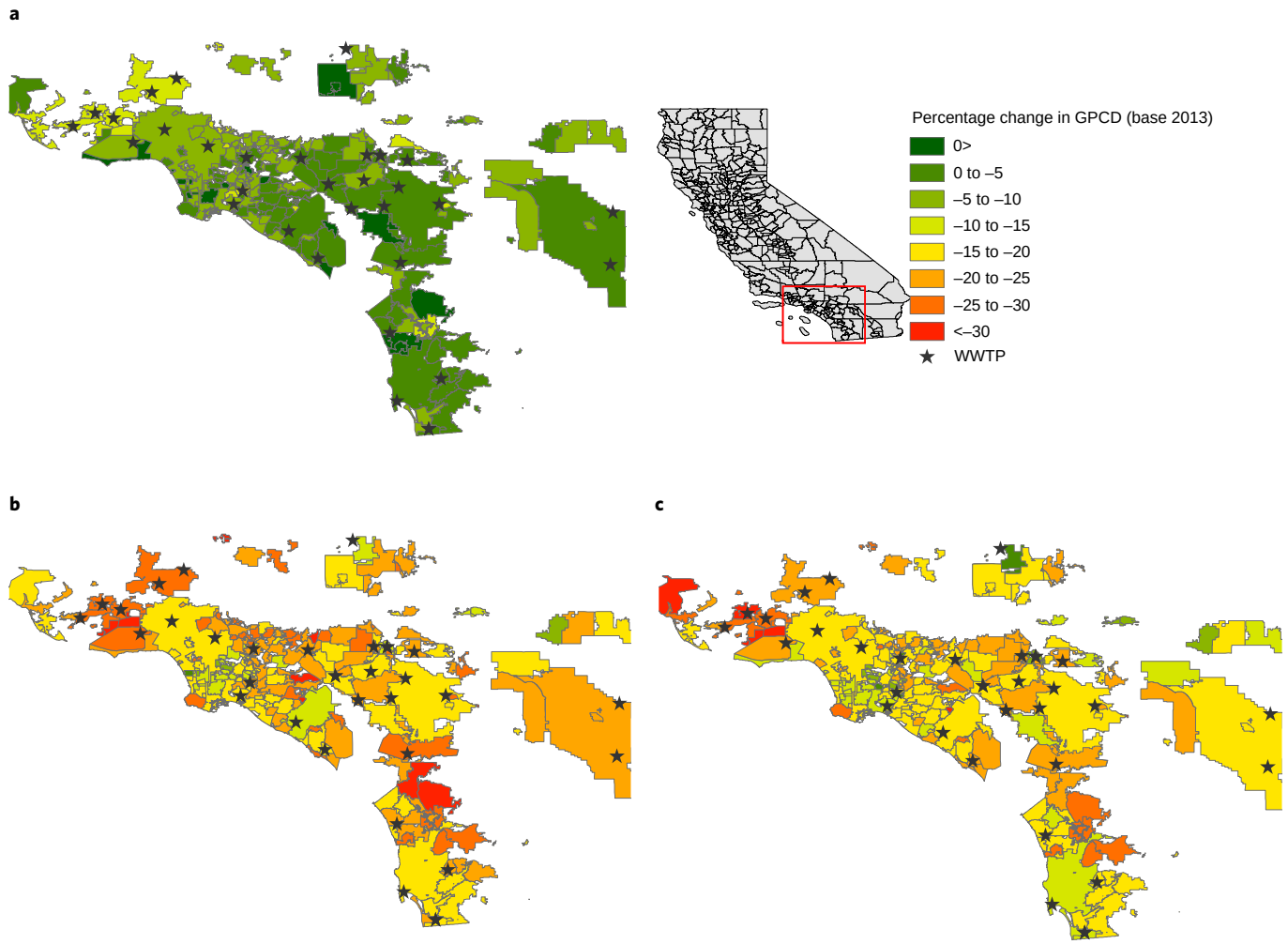


Fig. 2 | The locations of the sample WWTPs and corresponding water agency flow reductions. a, Voluntary (June 2014–May 2015) average change in GPCD (base 2013) = −6.56%. **b**, Mandate (June 2015–May 2016) average change in GPCD (base 2013) = −20.11%. **c**, Self-certification (June 2016–May 2017) average change in GPCD (base 2013) = −17.04%. The region shown in **a**–**c** is indicated by the red box at the top right, and the legend at the top right applies to **a**–**c**.

effluent TDS for each WWTP (Fig. 3). Our data (Fig. 3a) suggest that effluent flow and GPCD are positively related—the dashed line represents a fitted regression line with a positive slope from a simple ordinary least squares regression. As shown, most observations are in the third (negative–negative) quadrant, indicating that reductions in GPCD are most likely associated with reductions in effluent flow. This result is consistent with some fraction of conservation occurring due to reductions in indoor water use. In considering the colour of the dots—which correspond to the period in which reductions occurred—it seems that most of the observations during the voluntary period are centred around the origin (as indicated by the red dots), while there is an apparent shift in the number of observations to the third quadrant during the mandatory conservation period (as indicated by the green dots).

In general, we observed that GPCD is negatively related to effluent TDS concentrations (that is, declining GPCD and increased TDS; Fig. 3b), with both the majority of observations occurring in the second (negative–positive) quadrant, with a negative slope on the fitted regression line. These results—which suggest that greater reductions in GPCD lead to higher effluent TDS—are indicative of declining indoor use that results in more concentrated, or less diluted, wastewater. In regard to how the observations are concentrated throughout the different policy periods, we again see evidence of a stronger relationship between effluent TDS and GPCD under the conservation mandate than during the voluntary period (Fig. 3b).

Finally, we explored the general relationship between effluent flow and TDS (Fig. 3c). We observed that—although most of the observations are located in the second (negative–positive) quadrant, indicating that reductions in flow are associated with higher effluent TDS—the simple fitted regression line has a positive slope, suggesting that, as the percentage change in flow increases, the percentage change in TDS also increases. Relative to the previous two graphs (Fig. 3a,b), we did not observe much evidence that this relationship varied across the three policy periods. There seems to be confounding factors involved here that a simple two-way analysis cannot parse. For example, some agencies may draw on more saline source water during a drought than other agencies, or the wastewater-influent relationships may be impacted by seasonality—that is, stormwater runoff can infiltrate into wastewater collection pipes during wet periods.

To account for possible confounding effects that influence the relationship between effluent flow, effluent TDS and conservation policies, regression models with various fixed-effects specifications were used that control for monthly, seasonal, annual and WWTP invariant factors that may explain a substantial amount of variation

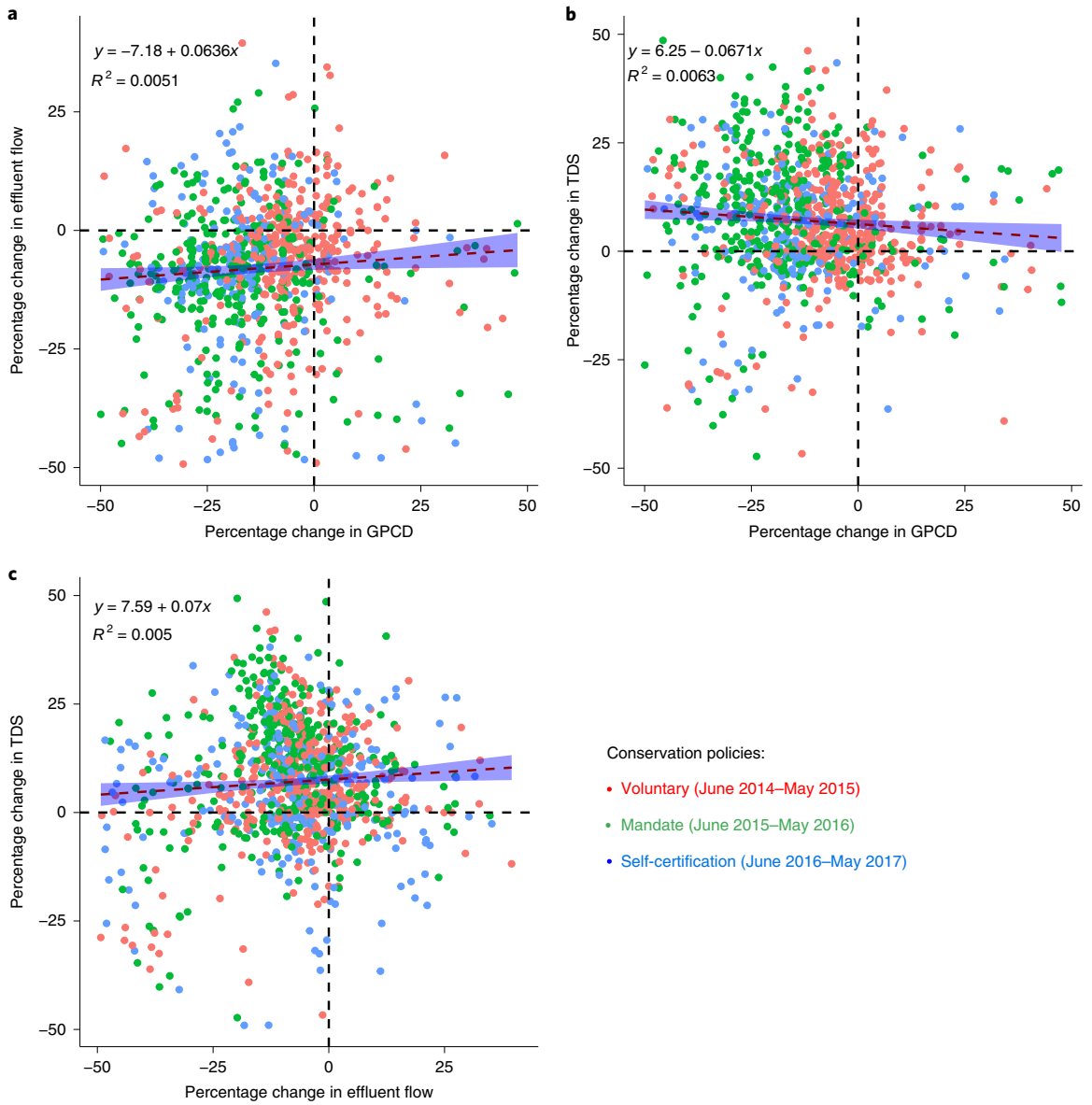


Fig. 3 | The monthly relationship between changes in effluent flow, GPCD and TDS. a–c, The monthly relationship between effluent flow versus GPCD (**a**), TDS versus GPCD (**b**) and TDS versus effluent flow (**c**) compared with the 2013 levels. The dashed red lines are the fitted values using a linear regression method along with the 95% confidence interval. The dashed black lines are the reference lines indicating zero on the *x* axis and *y* axis. All of the values are monthly level percentage changes compared with the same months in 2013 as a baseline. Note that negative values indicate reductions and positive values indicate increases relative to the 2013 levels.

that could otherwise bias the results. To this end, we developed a panel dataset of WWTP-level monthly effluent flow, effluent TDS and conservation-policy indicators. This panel begins January 2013 and ends in December 2017 and includes binary variables that capture periods representing the voluntary conservation policy (June 2014–May 2015), mandatory conservation policy (June 2015–May 2016) and the self-certification period following the end of the mandate (June 2016–May 2017); the dataset contains over 1,500 observations, including 33 WWTPs and three years of monthly data. The regression results measure the effect of conservation policies, controlling for WWTP characteristics (such as WWTP size) on effluent flow and TDS, as well as any seasonal, monthly or year-specific impacts on these variables. Further details and summary statistics of the data and variables used in this analysis are provided in the Methods and Supplementary Table 1.

Impacts of conservation policy on the effluent flow

Results from the fixed effects regression analyses are shown in Fig. 4. The red whisker plots summarize the relationship between the percentage change in monthly volume-weighted average effluent flow (natural-log- (ln)-transformed) and conservation policy for different specifications. Confidence intervals at the 95% level for all of the specifications are reported in the error bars and are clustered at the WWTP level to account for within-WWTP serial correlation in the error term[35]. Note that, as we have a semi-ln specification, the coefficients of the dummy variables cannot be interpreted as percentage changes. We applied the Halvorsen–Palmquist transformation to calculate the correct percentage reductions[36]. The three specifications include, as explanatory variables, ln of residential water consumption, WWTP fixed effects, season indicators and indicators for the three conservation policy periods. The second

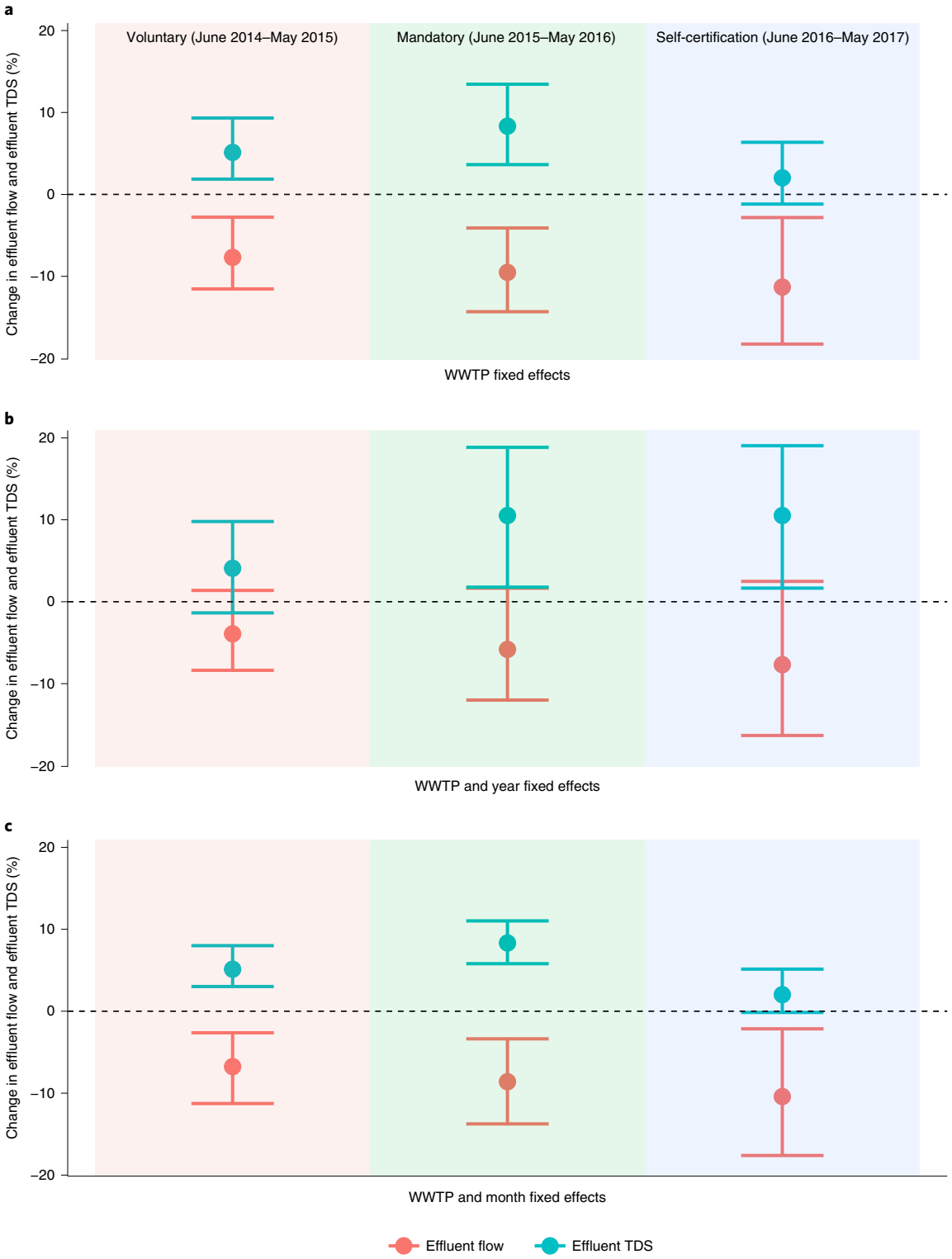


Fig. 4 | Analysis of the relationship between conservation policies and changes in effluent flow and TDS. a, Regression estimation results using a specification that controls for WWTP fixed effects. **b**, Regression results using WWTP and year fixed effects. **c**, Regression results using WWTP and month fixed effects. In each specification, we included additional controls. The full estimation results and clustered standard errors at the WWTP level for all of the specifications are provided in Supplementary Tables 2 and 3. The outcome of the misspecification test indicates that there is no evidence that the models are misspecified. For **a**–**c**, the error bars show the 95% confidence intervals.

specification includes year fixed effects (Fig. 4b), whereas the third specification replaces the year fixed effects with a monthly fixed effect (Fig. 4c). As shown, we found a negative and statistically significant effect from each of the conservation policies on effluent flow relative to the period before the conservation measures were enacted. Furthermore, spring (March–May), summer

(June–August) and autumn (September–November) flow rates are lower than winter (December–February; the default) flow rates, most likely due to the higher rains and consequent storm-water flows and/or groundwater infiltration that enter the effluent pipes during the winter. Additional specifications, estimated coefficients and details on the estimation results are provided in Supplementary Table 2.

Our results suggest that, even after controlling for unobservable differences in WWTPs (such as size and age) and both monthly and yearly time-invariant effects, water conservation measures over the last few years have resulted in a decrease in the volume of wastewater conveyed to WWTPs. More specifically, our results indicate that, on average, the voluntary conservation policy reduced effluent flow by 6.76% (0.41 million gallons per day), the mandatory conservation reduced effluent flow by 9.43% (0.57 million gallons per day) and the self-certification requirement reduced effluent flow by 10.42% (0.63 million gallons per day). Although the point estimates in Fig. 4 are shown to be larger during the conservation mandate and self-certification periods relative to the voluntary period, they are all within one another's error bounds. Moreover, these results indicate that each of the conservation policies reduced the amount of wastewater available for reuse, and such reductions may also impact streamflow and downstream water quality in cases in which the wastewater was discharged to a stream.

Results for robustness checks are provided in Supplementary Table 4. As shown in column 1 of Supplementary Table 4, we found that the relationship between conservation policy and effluent flow does not differ significantly between WWTPs that receive influent from water agencies that rely primarily on imported water versus water agencies that rely primarily on local sources for the voluntary and mandated conservation periods. We also observed (Supplementary Table 4, column 2) that dropping the WWTP that has RO technology does not lead to any appreciable change in our results. We also tested a general-to-specific statistical methodology as an alternative to the fixed effects and found no appreciable change in statistical significance levels or coefficient magnitudes (Supplementary Table 6).

Impacts of conservation policy on effluent TDS

The blue whisker plots in Fig. 4 show the regression results of the relationship between percentage change in monthly volume-weighted average TDS concentrations (ln-transformed) and conservation policies for different specifications. Similar to the effluent flow regressions, confidence intervals at the 95% confidence level for all of the specifications are shown as error bars and are clustered at the WWTP level. All three specifications shown in Fig. 4 include the ln of effluent flow, WWTP fixed effects, season indicators and indicators for the three conservation policy periods, with the second and third specifications adding a yearly and monthly fixed effect (similar to the effluent flow regressions).

The results suggest that there is a positive and statistically significant effect of conservation policies on effluent TDS relative to the baseline (2013) levels. More specifically, in the last specification (WWTP and monthly fixed effects), we found that voluntary, mandate and self-certification conservation policies were associated with increased TDS in effluent flow by 5.13%, 8.33% and 2.02%, respectively. The larger effect of the mandate on effluent flows relative to the voluntary and self-certification periods was statistically significant. To put these numbers in perspective, the average TDS level in 2013 (baseline period) in our sample is approximately 670 ppm. The total increases in TDS levels over the conservation periods are 34.33 ppm, 55.78 ppm and 13.53 ppm for voluntary, mandate and self-certification periods, respectively.

By performing an analysis using additional specifications, we found similar results. For example, we found an inverse and statistically significant relationship between effluent flow and effluent TDS in specifications that do not include WWTP-level fixed effects (Supplementary Table 3). Further robustness checks are presented in Supplementary Table 5. As shown, estimates are not sensitive to the water agencies' primary source water (local versus imported), and the estimated coefficients are robust when we add the WWTP with RO technology to the sample. We also applied the general-to-specific methodology to the data in place of the fixed effects and found no appreciable differences in the magnitude nor statistical significance of the estimated coefficients (Supplementary Table 6).

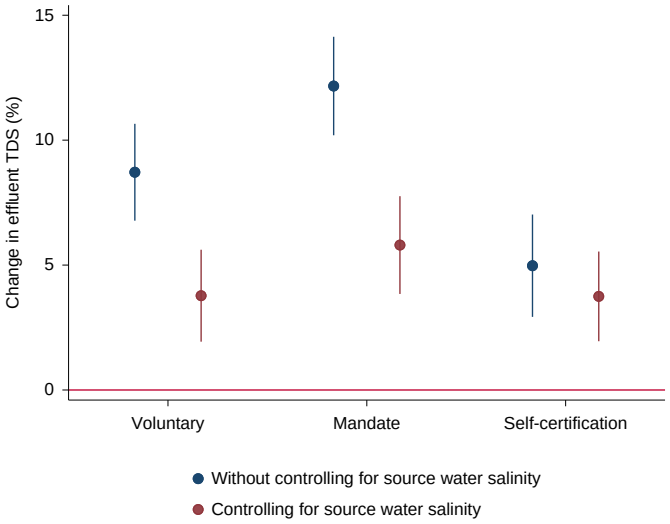


Fig. 5 | Analysis of the relationship between conservation policies and changes in effluent TDS with and without controlling for the salinity levels for the source water. Regression results using WWTP and month fixed effects. The full estimation results are reported in columns 3 and 4 of Supplementary Table 5. The subsample includes 11 WWTPs, in which we observed the salinity level of the source water. The whiskers show the 95% confidence intervals.

A time-variant effect that could bias the conservation policy-effluent salinity estimation is if there is a change in the source-water salinity. To investigate this potential bias, we collected data on source-water salinity levels for 11 WWTPs over the entire study period. Using this subsample of WWTPs, we estimated our preferred specification (regression with WWTP and calendar-month fixed effects). Results with and without controlling for the source-water salinity levels are presented in Fig. 5. As indicated, without controlling for source-water salinity levels, the estimated coefficients of the effect of conservation policies on effluent TDS would be overestimated. When we control for the source-water salinity levels, the estimated effects for voluntary, mandate and self-certification conservation policies drop from 8.72%, 12.17% and 4.97% to 3.77%, 5.80% and 3.75%, respectively. Further details on the estimated coefficients are provided in columns 3 and 4 of Supplementary Table 5. Here, it is important to recognize that—although our results still show that the conservation policies resulted in a spike in effluent salinity, the greatest being associated with the conservation mandate—effluent salinity may further increase or decrease depending on how the drought impacts source-water salinity.

Discussion and conclusions

We found that the recent conservation policies imposed on water agencies by the state reduced both the availability and quality of effluent flows—flows that are increasingly targeted to argument local water supplies, especially during drought. The results, which are consistent with basic water balance, suggest that policies that

lead to reductions in indoor water use will necessarily decrease the supply of water that is available for recycling, reuse or discharge to local streams/riverine systems and increase the constituent concentrations of the influent, notably salinity. The magnitude of such impacts to society will depend on many factors, including the extent to which water-reduction policies target outdoor versus indoor water efficiency and conservation, the ability by which WWTPs can respond to lower and more concentrated influent flows[34], the degree to which source-water salinity changes and the sensitivity of the services that receive TMW to changes in effluent flow and quality.

Our results contribute to a broader body of literature showing that the failure of policy to consider the interdependencies and coevolution of factors that comprise a human-water system will lead to unanticipated water scarcity and unintended consequences[19–25], consequences that will be borne by the environment, the users of recycled water, water managers and, therefore, society as a whole. Diligence in incorporating water balance is one ingredient to help to avoid such unanticipated and unintended outcomes, as is an understanding that lowering the costs of water use (for example, through improvements in technical efficiency) will often lead to production or behavioural changes that result in increased water use and greater levels of scarcity absent some sort of rationing. Recognition that the water system is not static but, rather, is comprised of demand and supply cycles that evolve over time and in response to changes in water availability is needed for effective water management. Although efforts to promote indoor water conservation and wastewater reuse are necessary tools to combat rising water scarcity and more frequent and intense drought, failure to consider the outcomes of this analysis and the evolving literature may lead to less resilient water systems that confront greater levels of water scarcity than is necessary.

Methods

Data. We used multiple data sources to evaluate this issue. Our data on WWTP characteristics (for example, effluent fl w and salinity, and capacity) were obtained primarily from the WWTPs or water agencies themselves. There are more than 300 WWTPs in Southern California. We dropped WWTPs that received influent from multiple water agencies because it was not feasible to determine the exact discharge volume and quality for each of those agencies (individually). Of the 109 WWTPs that remained, all of which were contacted; 45 treatment plants responded to our request for data. Our request generated data on monthly fl w and TDS from 2013–2017 along with additional information on WWTP characteristics, such as capacity. Specifi ally, the data that we requested and were provided included (1) monthly effluent volume of wastewater (monthly average) at each plant starting from 2013–2017 (note that effluent fl w represents the volume of wastewater discharged from the treatment plants) and (2) monthly average effluent TDS from 2013–2017 at each plant (which captures the concentration of salt or TDS in the wastewater effluent).

To control for changes in water use, we merged the WWTP dataset with State Water Resources Control Board (SWRCB) reports at the water-agency level to obtain the total monthly single-family residential water consumption and average monthly gallons per capita per day (GPCD). After merging and cleaning these two datasets, we had a complete dataset for 34 WWTPs. However, one of the WWTPs was equipped with RO technology, and can treat higher TDS levels. For the purpose of this research, we eliminated this WWTP from the effluent TDS analysis, leaving us with data on 34 WWTPs for the effluent flow analysis and 33 WWTPs for the effluent TDS analysis; these datasets formed the sample for our analysis. Definitions, details and summary statistics on the variables used in this paper are provided in Supplementary Table 1. As indicated in Supplementary Table 1, on average, our sample effluent flow is 12.21 million gallons per day, with average TDS levels of 788.51 ppm, and with an average design flow of 25.02 million gallons per day. Note that we were able to collect information on source-water salinity levels for only 11 WWTPs from 2013–2017. This information merged with our primary dataset and enabled us to run additional specifications while controlling for source-water salinity levels.

We encountered a few challenges in our data collection efforts. First, WWTP effluent flow rates vary throughout the year, depending on seasonal storm runoff and infiltration. Along the California coast, the lowest flow rates (or dry weather flow rates) typically occur from July to September. We therefore converted the daily flow rates collected from the SWRCB website and WWTPs to monthly averages to better understand these variations. Nearly all of the WWTPs measure the volume of wastewater for effluent on a daily basis. However, in some cases, there are data gaps due to seasonal monitoring or alterations to permits throughout 2013–2017. This was an issue, in particular, for obtaining TDS data. During data collection, some of the treatment plants mentioned that they are not required to test for TDS. For example, we were informed that a WWTP that discharges effluent into the ocean does not, and is not required to, measure its TDS. Furthermore, some of the WWTPs started to measure their salinity level only after the governor's mandated conservation declaration in June 2015, which was problematic because our analysis compares effluent quality under the three different conservation policies from 2014 to 2017 to baseline (2013) estimates (before and after conversation policies went into effect) to measure the difference over the drought period. As a consequence, our dataset includes only WWTPs for which we were able to collect monthly data on effluent flow and TDS.

Statistical analysis. The primary challenge for estimating the effect of water conservation policies on effluent flow and salinity is that there may be unobservable factors that are related to both the water policies enacted during the drought (our independent/explanatory variables of interest) and effluent flow and salinity levels (our dependent variables of interest). Such unobservables, if unaccounted for, would probably lead to omitted-variable bias in our estimator for the effect of the policies on effluent flow and effluent salinity. The rich panel dataset of monthly WWTP-level effluent flow and salinity enabled us to include various fixed effects and control for such unobservable factors. Fixed-effects estimation is a well-accepted and robust approach to control for correlated time-invariant unobservable factors that vary across categories (for example, technology), thereby enabling the researcher to reduce potential bias and identify causality[37–39]. In our preferred specification, we controlled for WWTP and calendar-month fixed effects, where the WWTP fixed effects were used to control for time-invariant WWTP characteristics, such as location, size, technology and management strategies. In addition to WWTP fixed effects, by adding calendar-month fixed effects, we controlled for factors that are common to all WWTPs within a given calendar month, such as seasonality of effluent flow and salinity.

Specifically, we estimated a fixed-effects model with WWTPs, month and year fixed effects using monthly level data. First, we used the following equation to establish a relationship between water conservation policies and effluent flow from the WWTPs.

$$\ln(\text{Effluent}_{imt}) = \beta_1 \ln(\text{MPW}_{imt}) + \beta_2 \text{VCon}_{mt} + \beta_3 \text{MCon}_{mt} + \beta_4 \text{SelfCer}_{mt} + \text{WWTP}_i + u_{imt} \quad (1)$$

where Effluent_{imt} is the monthly average effluent flow from the WWTP i in month m and year t measured as millions of gallons per day; MPW_{imt} is the average monthly water deliveries to single-family residential accounts in the WWTP i service area measured as gallons per day. VCon_{mt} is an indicator variable that takes 1 for the voluntary conservation period (June 2014–May 2015) and 0 otherwise; MCon_{mt} is an indicator variable for the mandatory conservation period (June 2015–May 2016); and SelfCer_{mt} is an indicator variable for the self-certification period following the end of the mandate (June 2016–May 2017). WWTP_i is a WWTP service area fixed effect, and u_{imt} captures all unobserved factors affecting the dependent variable. We also estimated additional specifications using month and year fixed effects.

Next, we used equation (2) to investigate the relationship between effluent TDS and water conservation policies.

$$\ln(\text{TDS}_{imt}) = \beta_1 \ln(\text{Effluent}_{imt}) + \beta_2 \text{VCon}_{mt} + \beta_3 \text{MCon}_{mt} + \beta_4 \text{SelfCer}_{mt} + \text{WWTP}_i + u_{imt} \quad (2)$$

TDS_{imt} is the monthly volume-weighted average effluent TDS in the WWTP i in month m and year t measured as ppm; the other variables are defined as in equation (1). Similar to the previous equation, we also estimated additional specifications using month and year fixed effects.

Data availability

All data that support the findings of this study are available from the corresponding authors on request.

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Author contributions

K.S., M.N. and D.J. designed research. K.S., M.N., R.A. and Q.T. performed research. K.S., M.N., R.A., Q.T. and D.J. wrote the paper.

Competing interests

The authors declare no competing interests.

Additional information

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